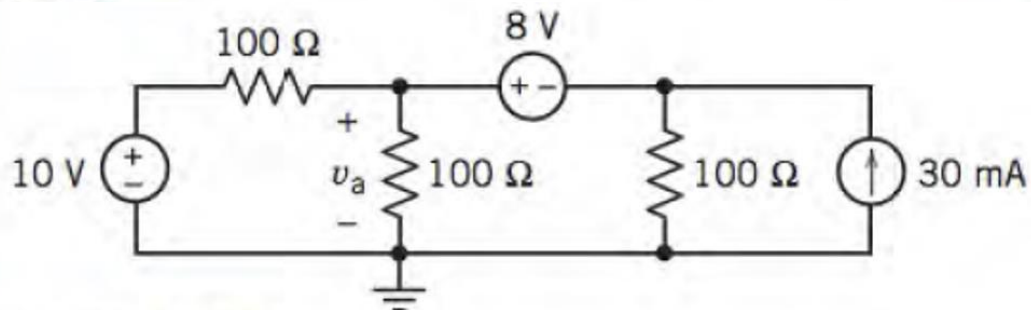
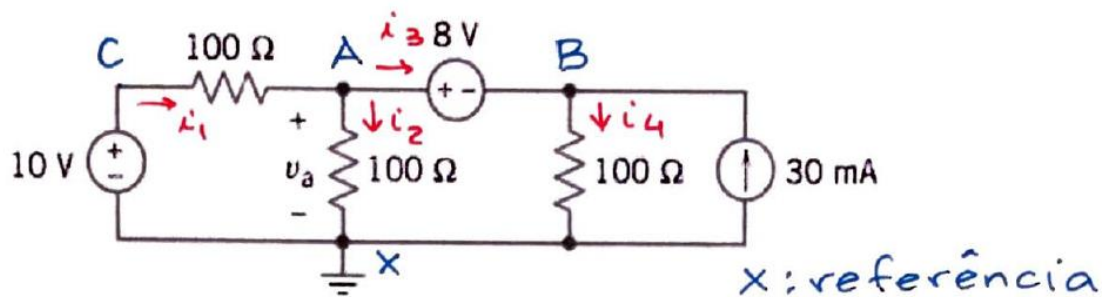


Análise Nodal – Lista de Exercícios 3* RESOLUÇÃO

P 4.3-3 Determine a tensão de nó v_a no circuito da Figura P 4.3-3.



SOLUÇÃO



nó C:

$$v_C = 10V$$

supernó A-B:

$$i_1 + 0,03 = i_2 + i_4$$

$$\frac{v_C - v_A}{100} + 0,03 = \frac{v_A}{100} + \frac{v_B}{100}$$

$$v_C - v_A + 3 = v_A + v_B$$

$$2v_A + v_B = v_C + 3$$

$$\Rightarrow \boxed{2v_A + v_B = 13}$$

fonte 8V:

$$\boxed{v_A - v_B = 8}$$

$$\begin{cases} 2v_A + v_B = 13 \\ v_A - v_B = 8 \end{cases}$$

$$3v_A = 21 \Rightarrow \boxed{v_A = 7\text{ V}}$$

$$\boxed{v_B = -1\text{ V}}$$

tensões nodais:

$$v_A = 7\text{ V}$$

$$v_B = -1\text{ V}$$

$$v_C = 10\text{ V}$$

correntes:

$$i_1 = \frac{v_C - v_A}{100} = \frac{10 - 7}{100} = \frac{3}{100} \Rightarrow i_1 = 0,03\text{ A}$$

$$i_1 = 30\text{ mA}$$

$$i_2 = \frac{v_A}{100} = \frac{7}{100} \Rightarrow i_2 = 0,07\text{ A}$$

$$\Rightarrow i_2 = 70\text{ mA}$$

$$i_4 = \frac{v_B}{100} = \frac{-1}{100} \Rightarrow i_4 = -0,01\text{ A}$$

$$\Rightarrow i_4 = -10\text{ mA}$$

i_3 :

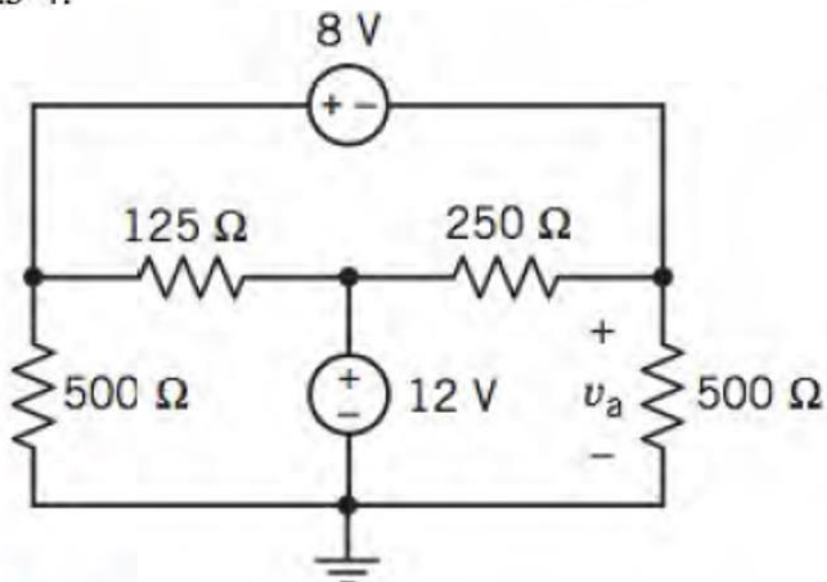
$$i_1 = i_2 + i_3 \quad (\text{nó A})$$

$$\Rightarrow i_3 = i_1 - i_2 = 30 - 70 \Rightarrow i_3 = -40\text{ mA}$$

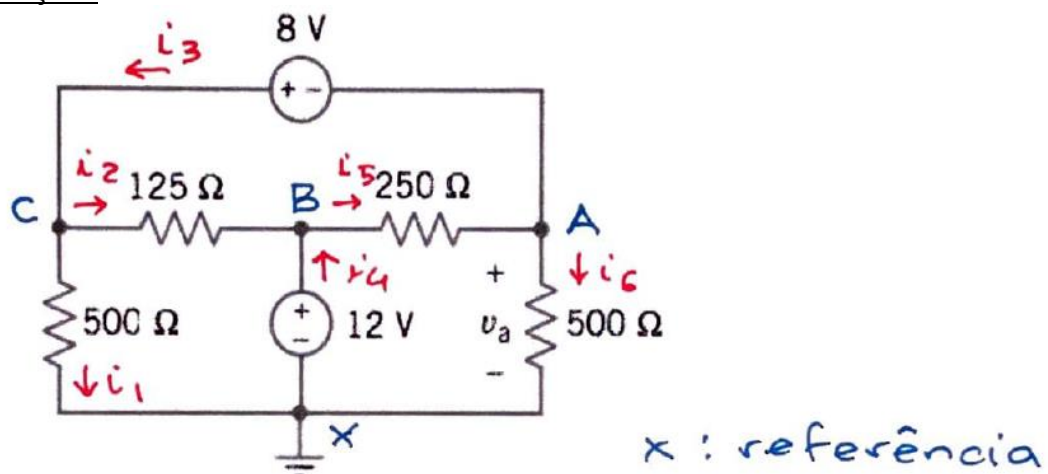
$$i_3 + 30 = i_4 \quad (\text{nó B})$$

$$\Rightarrow i_3 = i_4 - 30 = -10 - 30 \Rightarrow i_3 = -40\text{ mA}$$

P 4.3-4 Determine a tensão de nó v_a no circuito da Figura P 4.3-4.



SOLUÇÃO



nó B:
$$v_B = 12 \text{ V}$$

fonte 8V:
$$v_C - v_A = 8$$

supernd' A-C:

$$i_1 + i_2 + i_6 = i_5$$

$$\frac{v_c}{500} + \frac{v_c - v_B}{125} + \frac{v_A}{500} = \frac{v_B - v_A}{250}$$

$$2v_c + 8v_c + 8v_B + 2v_A = 4v_B - 4v_A$$

$$6v_A + 10v_c = 12v_B$$

$$\Rightarrow \boxed{6v_A + 10v_c = 144}$$

$$\begin{cases} -v_A + v_c = 8 \xrightarrow{\times 6} -6v_A + 6v_c = 48 \\ 6v_A + 10v_c = 144 \end{cases}$$

$$\Rightarrow 16v_c = 192 \Rightarrow \boxed{v_c = 12V}$$

$$v_A = v_c - 8 \Rightarrow \boxed{v_A = 4V}$$

tensões nodais:

$$v_A = 4V$$

$$v_B = 12V$$

$$v_c = 12V$$

corrientes:

$$i_1 = \frac{U_C}{500} = \frac{12}{500} \Rightarrow i_1 = 0,024 \text{ A}$$

$$i_1 = 24 \text{ mA}$$

$$i_2 = \frac{U_C - U_B}{125} = \frac{12 - 12}{125} \Rightarrow i_2 = 0$$

$$i_5 = \frac{U_B - U_A}{250} = \frac{12 - 4}{250} \Rightarrow i_5 = 0,032 \text{ A}$$

$$i_5 = 32 \text{ mA}$$

$$i_6 = \frac{U_A}{500} = \frac{4}{500} \Rightarrow i_6 = 0,008 \text{ A}$$

$$i_6 = 8 \text{ mA}$$

i_3 :

$$\text{no}'C: i_1 + i_2 = i_3 \Rightarrow i_3 = 24 \text{ mA}$$

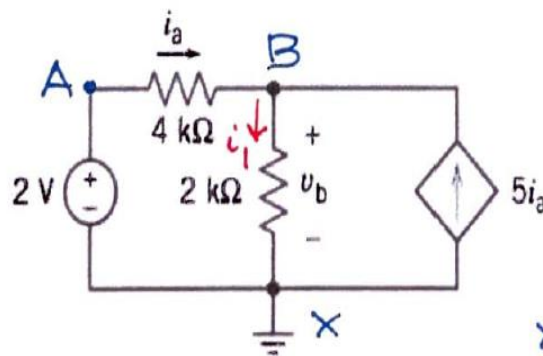
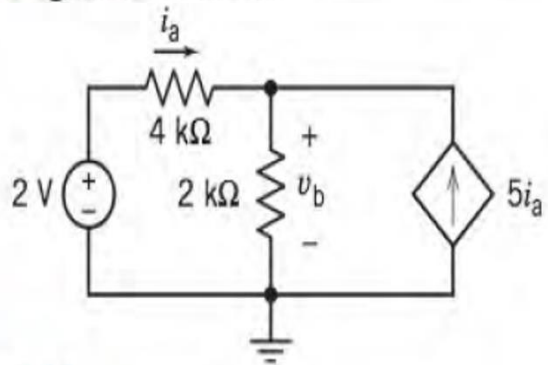
$$\text{no}'A: i_3 + i_6 = i_5 \Rightarrow i_3 = 24 \text{ mA}$$

i_4 :

$$\text{no}'B: i_4 + i_2 = i_5 \Rightarrow i_4 = 32 \text{ mA}$$

$$\text{no}'X: i_4 = i_1 + i_6 \Rightarrow i_4 = 32 \text{ mA}$$

P 4.4-3 Determine o valor da tensão de nó v_b no circuito da Figura P 4.4-3.



X: referência

SOLUÇÃO

nó A: $\boxed{v_A = 2\text{ V}}$

nó B:

$$i_a + 5i_a = i_1$$

$$6i_a = i_1$$

$$i_a = \frac{v_A - v_B}{4}$$

$$\Rightarrow 6 \left(\frac{v_A - v_B}{4} \right) = \frac{v_B}{2}$$

$$3v_A - 3v_B = v_B$$

$$4v_B = 3v_A$$

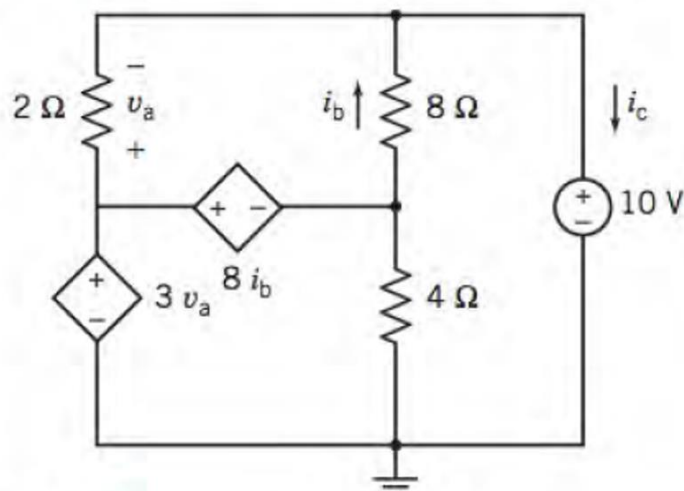
$$4v_B = 6 \Rightarrow \boxed{v_B = 1,5\text{ V}}$$

correntes:

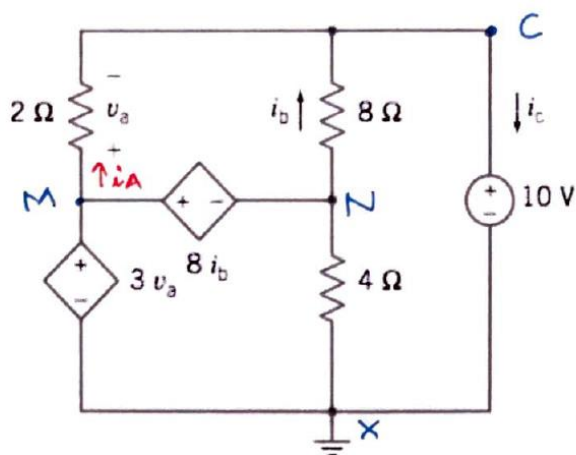
$$i_1 = \frac{v_B}{2} \Rightarrow i_1 = 0,75\text{ mA}$$

$$i_a = \frac{v_A - v_B}{4} \Rightarrow i_a = 0,125\text{ mA}$$

P 4.4-7 Determine o valor da corrente i_c na Figura P 4.4-7.



SOLUÇÃO



x: referência

nó' M:

$$v_M = 3v_A$$

$$v_A = v_M - v_C$$

$$v_M = 3(v_M - v_C)$$

$$v_M = 3v_M - 3v_C$$

$$\Rightarrow \boxed{2v_M - 3v_C = 0}$$

nó' C:

$$\boxed{v_C = 10V}$$

$$\Rightarrow 2v_M = 3v_C$$

$$2v_M = 30$$

$$\Rightarrow \boxed{v_M = 15V}$$

fonte $8i_B$:

$$v_M - v_N = 8i_B$$

$$i_B = \frac{v_N - v_C}{8}$$

$$\Rightarrow v_M - v_N = 8 \left(\frac{v_N - v_C}{8} \right)$$

$$v_M - v_N = v_N - v_C$$

$$\Rightarrow 2v_N = v_M + v_C$$

$$2v_N = 10 + 15$$

$$2v_N = 25 \quad \Rightarrow \quad \boxed{v_N = 12,5V}$$

tensões nodais:

$$v_M = 15V$$

$$v_N = 12,5V$$

$$v_C = 10V$$

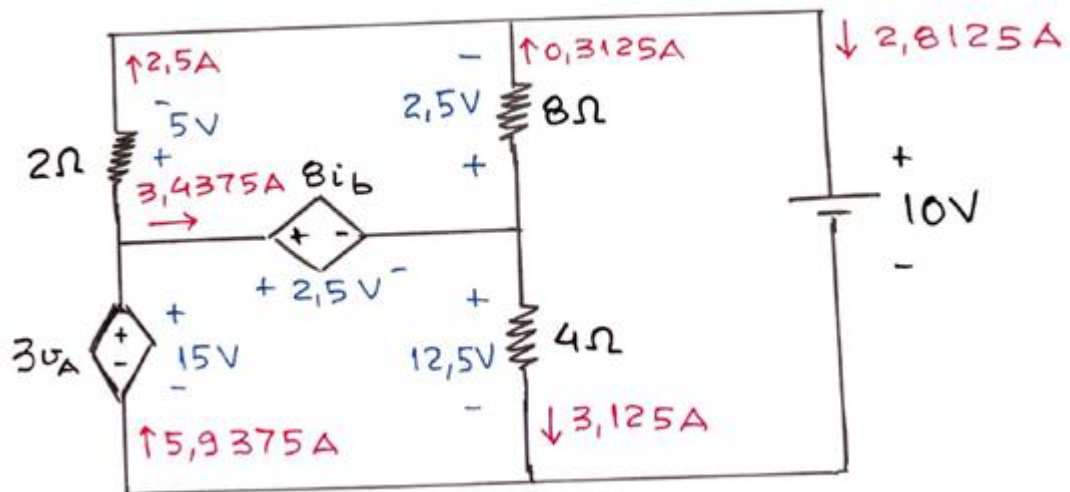
$$i_C = i_A + i_B$$

$$i_A = \frac{v_M - v_C}{2} = \frac{15 - 10}{2} \Rightarrow i_A = 2,5A$$

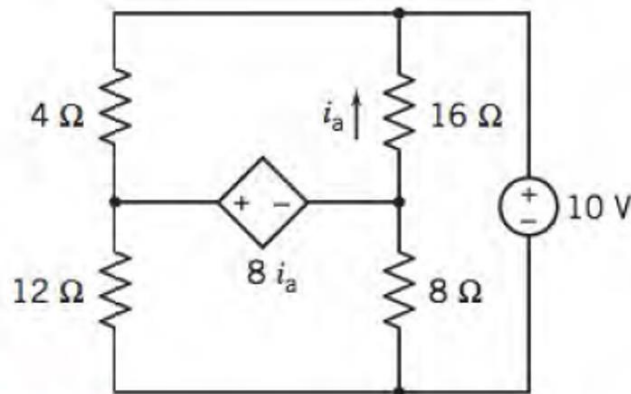
$$i_B = \frac{v_N - v_C}{8} = \frac{12,5 - 10}{8} \Rightarrow i_B = 0,3125A$$

$$\Rightarrow i_C = 2,5 + 0,3125$$

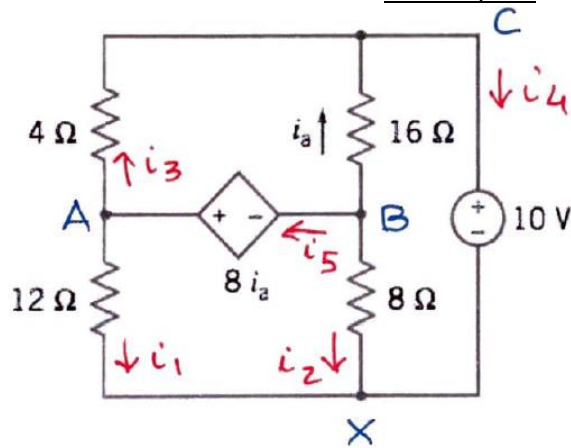
$$\Rightarrow \boxed{i_C = 2,8125A}$$



P 4.4-8 Determine o valor da potência fornecida pela fonte dependente na Figura P 4.4-8.



SOLUÇÃO



x: referência

nó C:
$$v_C = 10\text{ V}$$

fonte dependente:

$$v_A - v_B = 8i_A$$

$$i_A = \frac{v_B - v_C}{16}$$

$$v_A - v_B = 8 \left(\frac{v_B - v_C}{16} \right)$$

$$2v_A - 2v_B = v_B - v_C$$

$$2v_A - 3v_B = -v_C$$

$$\Rightarrow \boxed{2v_A - 3v_B = -10}$$

supernó A-B:

$$i_1 + i_3 + i_A + i_2 = 0$$

$$\frac{v_A}{12} + \frac{v_A - v_C}{4} + \frac{v_B - v_C}{16} + \frac{v_B}{8} = 0$$

$$4v_A + 12v_A - 12v_C + 3v_B - 3v_C + 6v_B = 0$$

$$\Rightarrow 16v_A + 9v_B = 15v_C$$

$$\Rightarrow \boxed{16v_A + 9v_B = 150}$$

$$\begin{cases} 2v_A - 3v_B = -10 \xrightarrow{\times 3} 6v_A - 9v_B = -30 \\ 16v_A + 9v_B = 150 \end{cases}$$

$$\Rightarrow 22v_A = 120 \Rightarrow \boxed{v_A = 5,45 \text{ V}}$$

$$\Rightarrow \boxed{v_B = 6,97 \text{ V}}$$

corrientes:

$$i_1 = \frac{v_A}{12} = \frac{5,45}{12} \Rightarrow i_1 = 0,45 \text{ A}$$

$$i_2 = \frac{v_B}{8} = \frac{6,97}{8} \Rightarrow i_2 = 0,87 \text{ A}$$

$$i_3 = \frac{v_A - v_C}{4} = \frac{5,45 - 10}{4} \Rightarrow i_3 = -1,1375 \text{ A}$$

$$i_A = \frac{v_B - v_C}{16} = \frac{6,97 - 10}{16} \Rightarrow i_A = -0,1894 \text{ A}$$

i_5 :

$$\text{nó A: } i_5 = i_3 + i_1$$

$$i_5 = -1,1375 + 0,45$$

$$\Rightarrow i_5 = -0,6875 \text{ A}$$

$$\text{nó B: } i_5 + i_A + i_2 = 0$$

$$i_5 = -i_A - i_2$$

$$i_5 = 0,1894 - 0,87$$

$$\Rightarrow i_5 = -0,6806 \text{ A}$$

$$\Rightarrow i_5 = -0,68 \text{ A}$$

fonte dependente:

$$\text{corrente: } i_5 = -0,68 \text{ A}$$

$$\begin{aligned} \text{tensão: } 8i_A &= 8 \times (-0,1894) \\ &= -1,5152 \text{ V} \end{aligned}$$

potência:

$$P_f = 1,5152 \times 0,68$$

$$\Rightarrow \boxed{P_f = 1,03 \text{ W}}$$

*Fonte: Richard C. Dorf, James A. Svoboda **Introdução aos Circuitos Elétricos**